

WHITE PAPER · DEVELOPERS & RESEARCHERS

Toddler a governed, specialisable learning agent

A governed learning agent, its path to useful digital and physical work, PLS/PAR evidence, and a five-year business scenario.

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Version 0.2 (work-path revision) · 2026.10

github.com/virtuanalytica/toddler

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| Summary

Toddler is a learning agent built to be the **general base that every deployment specialises**: a developer or consultant adds company guardrails, rule questions and expert models to reach one role in an organisation's function house, but can never remove the base safety.

Key points

- Decisions maximise one utility (task reward, weekly profit, energy and compute cost, expected risk) *subject to* hard STOP rules, so no reward can buy a forbidden action.
- Every logic step of a neonatal brain-network study (Wee et al., 2017), 25 in total, is linked to a Toddler function, and all 25 resolve to code in a woven knowledge graph with zero coverage gaps.
- The evaluation stack reproduces the study's analysis pipeline on public data: consensus clustering with a degree-preserving null model ($z = 7.6$, $p = 0.001$ on Zachary's karate club) and nested 10×10 cross-validation (accuracy 0.912, sensitivity 0.944, specificity 0.858 on the Wisconsin breast-cancer data).
- Status update, 11 October 2026: five G4-search children survived two private 30-seed navigation trials and a verified lineage import. The cognitive curriculum, general agent work and physical deployment have separate evidence gates; this result proves none of those broader claims.

Questions this paper answers

How should an agent weigh learning signal, business value and energy without letting any of them override safety? What can the structure of an infant's brain network teach about evaluating an agent early in its life? How can one base be specialised per company role while staying auditable?

| Problem and design goals

"Toddler" is not a formal term in neural-network literature but a common metaphor for a network early in training, and the toddler is the usual benchmark for efficient learning.

A human toddler recognises a new animal after a few examples and generalises it to toys and cartoons, while current networks often need thousands to millions of examples; the human brain runs on about 20 W against roughly 1 MW for AlphaGo (Salelanonda, Embedded.com). Toddler takes three design goals from that comparison and adds two from deployment practice:

Goal	Design consequence
Learn from few, verified signals	Reward comes only from an independent judge; claiming "done" without confirmation is the worst outcome
Generalise safely	Hard STOP constraints outside the utility; low confidence hands off to a human
Be energy-aware	Joules and euros are explicit terms in every decision and in expert routing
One base, many roles	Declarative specialisation that can only add constraints
Provable behaviour	Hash-chained audit trail with scoped, sanitised views

Physical care is safety-critical (EU AI Act high-risk, Medical Device Regulation, Machinery Regulation). Physical contact with infants is out of scope by design; elderly care starts with non-contact help and always with a caregiver present.

03 · ARCHITECTURE

| Architecture and objective

The brain has four layers that run in a fixed order: a deterministic logic layer that can veto, a fast reflex path, a slow deliberate path, and a decision step that maximises utility under the veto.

inputs (web now, sensors later)	
-> logic layer: STOP rules, allowlist, provenance	[veto]
-> fast path: Jev physical-rule probabilities	[reflex, <= 20 ms]
-> slow path: planner + world model + memory graph	[deliberate]
-> choose a maximising $U(a)$ under STOP	[decision]
-> act locally, or relay to a knitweb peer (Pulse)	[execution]
-> judge confirms -> reward; weekly -> profit	[learning signal]
-> append to the hash-chained audit trail	[oversight]

Figure 1. Toddler's processing order. Safety runs first and can veto everything after it.

The objective

```

U(a) = E[reward(a)] + w_profit * E[profit(a)]
      - w_energy * energy_J(a) - w_compute * cost_EUR(a)
      - w_risk * P(failure | a) * severity(a)
subject to  STOP(a) == false

```

Four quantities are kept apart because they answer different questions. The loss function is what a learned component minimises during training; it shapes weights and is not the decision rule. The reward is the per-task learning signal: +1 for a judge-confirmed result, +0.2 for an honest "I don't know", +0.1 for new useful knowledge, -1 for a wrong result and -5 for claiming success without confirmation, minus cost. The profit is the weekly account (value minus energy, compute, human review time and expected incidents) and decides whether a phase continues. Energy and compute enter every decision.

STOP rules are constraints rather than penalties. A penalty invites the agent to compute whether a violation "pays"; a constraint removes the option. In the implementation (`objective.choose`), a candidate with reward 10^9 that would reveal a secret loses to a safe candidate with reward 0.1, and when every option is vetoed the function returns no choice so the caller hands off to a human.

Modules in version 0.1

Module	Responsibility
stop	Hard STOP rules; veto before utility
objective	Reward, profit, energy/compute cost, decision rule
structure	Module-graph analysis after Wee et al. (rows 6-13)
evaluation	Behaviour evaluation statistics (rows 2-4, 14-18, 23-24)
fastpath	Reflex layer: hard limits over Jev probabilities
provenance	Source and consent register
relay	Pulse relay to knitweb peers with verified cost
specialize	Roles, company guardrails, Jev rules, expert routing
audit	Hash-chained trail, prune anchor, scoped views
config + GUI	Validated deployment configuration (toddler-config/v1)

From infant brain networks to agent evaluation

Wee et al. (2017) built anatomical brain networks for 120 newborns from diffusion tensor imaging, grouped them by network structure, and found that the structure at birth was associated with behaviour at 24 and 48 months.

The study concerns real infant brains, not artificial networks. We use it as a [methodological template](#) for a question every agent operator faces: does the structure an agent starts with predict how it behaves later, and how would we know? Four groups emerged (sizes 22, 37, 44 and 17); modularity $Q = 0.293$ lay 45 standard deviations above 10,000 random graphs; a linear SVM separated the two extreme groups with 89.4 % accuracy (sensitivity 0.954, specificity 0.765); the right amygdala's clustering coefficient was linked to behaviour at 24 months and the right inferior frontal gyrus and insula to externalising behaviour at 48 months.

Complete mapping

#	Step in the study	Toddler function (module)
1	Cohort with ethics approval, consent, inclusion criteria	Source/consent register; no synthetic data (provenance)
2	Independent rater, age-standardised T-scores	Judge-only scoring, T-scores per task type (evaluation)
3	Internalising vs externalising axes	Inward (over-cautious) vs outward (overreach) rates (evaluation)
4	Two horizons; attrition reported	Task reward and weekly profit; missingness reported (evaluation, objective)
5	No sedation, clinician present, vitals monitored	Supervisor and hard limits for risky steps (fastpath, stop)
6	Registration to a 64-region atlas	Fixed capability atlas of 17 modules (structure)
7	Tractography with stopping criteria	Path tracing with explicit stop criteria (structure, fastpath)
8	Volume normalisation; sign test removes noise edges	<code>normalise_by_size</code> , <code>sign_test_edges</code> (structure)
9	Clustering coefficient per region	<code>clustering_profile</code> (structure)
10	Subject similarity, trimmed at $r = 0.60$	<code>similarity_matrix</code> , <code>isolate-free</code> <code>trim_threshold</code> (structure)
11	Louvain consensus over 10,000 runs	<code>consensus_partition</code> (structure)
12	Null model of 10,000 random graphs	<code>modularity_null_test</code> (structure)
13	Assignment confidence	<code>assignment_confidence</code> ; low $\rightarrow$ ask a human (structure, objective)
14	Extremes compared with Welch tests	<code>compare_extremes</code> (evaluation)
15	Nested 10-fold CV, mRMR, SVM-RFE, >65 % selection	<code>nested_feature_selection</code> (evaluation)
16	Sensitivity and specificity reported apart	<code>sens_spec</code> ; alarms count misses and false alarms (evaluation)
17	CCA with Bonferroni correction	<code>cca_test</code> (evaluation)
18	Correlations for visualisation only	Findings flagged exploratory or confirmatory (evaluation)

#	Step in the study	Toddler function (module)
19	Right amygdala (threat) predicts early behaviour	Threat detector first and best connected (fastpath)
20	Inferior frontal gyrus (inhibition), insula	STOP layer wired to every action (stop, fastpath)
21	Sensory regions early, higher regions later	Phase order: perception and safety before control (fastpath)
22	Prenatal environment shapes the amygdala	Curated data; scraped content is data, never instructions (provenance)
23	Robust after removing extremes	<code>robust_without_extremes</code> (evaluation)
24	Weak early signal; replication needed	Promotion only after confirmatory replication (evaluation)
25	Structure at birth predicts later behaviour	Governance shipped before capability (stop, objective)

Reproduction on real data

We ran the clustering pipeline on Zachary's karate club (34 members, an observed social network with a documented split). Consensus clustering over 1,000 Louvain runs produced 4 groups with [purity 0.941](#) against the two real factions; mean modularity $Q = 0.416$ against a degree-preserving null (double edge swaps, weights permuted) with mean 0.293 gave $z = 7.6$ and $p = 0.001$ over 1,000 null graphs, the smallest p-value 1,000 null graphs can resolve; 91 % of members were assigned with confidence above 0.75. The nested 10×10 selection on the first ten Wisconsin breast-cancer features reached accuracy 0.912, sensitivity 0.944 and specificity 0.858, and kept two features above the 65 % selection threshold (mean perimeter, mean concave points). All figures are regenerated by `scripts/measure_whitepaper_figures.py` into `docs/whitepaper/figures.json`.

Our findings suggested that neural organization established during fetal development could to some extent predict individual differences in behavioral profiles.

- Wee et al., Human Brain Mapping 38(3), 2017

| Specialisation, guardrails and audit

A deployment is a declarative file (`toddler-config/v1`) edited in the Toddler configurator and validated by Python, which stays the authority even if a modified GUI produced the file.

Extension points

- Role: a position in the function house (family, level, tasks) or a single task.
- Company guardrails: STOP rules owned by the company, e.g. a compliance officer's "never export personal data"; ids are namespaced `company:*` and the owner appears in every veto.
- Jev rules: extra typed questions for the reflex layer with slow and stop thresholds ($0 \leq \text{slow} \leq \text{stop} \leq 1$).
- Experts: helper models on the mixture-of-models base, each with measured quality, euro and joule cost per 1,000 tokens and an evidence reference; the router picks the expert whose quality minus cost beats the base model.

Safety is monotonic: `specialize.build` rejects any rule id that shadows a base rule and any question that replaces a base reflex question, so a specialisation can add constraints but never weaken them.

Audit trail

The trail follows the design `fillslava` built for `ClaudeClaw`: each row stores the previous hash and its own hash, so editing any row breaks verification from that row on; retention pruning moves a stored anchor so the remaining chain still verifies; secret-shaped values are refused at write time. Readers get scoped views: a compliance audience sees guardrail and STOP events, a deployer only the events of their own role, with e-mail and IP masked and every row still carrying its hash for independent checking.

P2P compute, credentials and the physical roadmap

Pulse relay over knitweb

Pulse is a trustless relay to compute with node-to-node sharing. Toddler reuses knitweb's proof-of-useful-work primitives: a peer's result is accepted only after sampled re-execution, and the sample size comes from knitweb's exact hypergeometric `required_samples`. For a job of 1,000 output blocks and a tolerated 1 % miss of a 1 % corruption, 368 blocks are re-run. With electricity at €0.30 per kWh and a local job of 36 MJ, local execution costs €3.00 while a peer at €1.00 plus verification costs €2.10; a peer whose collateral makes fraud profitable is refused, and peers above 90 % load are skipped. The woven knowledge graph (pull request #5) compiles to a deterministic knitweb synaptic bundle whose round trip is checked by that pull request's tests; until the operator signs it, it is written as an unsigned file that must not be shared.

Credentials and multiple identities

A toddler has several e-mail identities: one per provider, so a revoked key never exposes another account. Keys are stored in OpenBao under `toddler/<identity>/<provider>/<key>`. Creation uses only official endpoints (Google API Keys, Cloudflare tokens, OpenAI service accounts, GitHub App installation tokens, AWS IAM) behind the operator-approval STOP rule; Anthropic and TransIP do not allow key creation by API and become human tasks. Rotation never revokes the old key before the new one is verified. Mass account creation, CAPTCHA bypass and temp-mail farming are out of scope. This module is written and tested locally but is not in the repository; it is pushed only after explicit operator approval.

Physical roadmap

The physical Toddler combines camera and lidar perception, industrial-grade motion precision (ISO 10218, ISO/TS 15066 contact limits), learned policies in the style of current legged and humanoid platforms trained in simulation, and Jev answering many physical-rule questions in parallel for the reflex layer. The reflex is already implemented in software: a measured force, speed or proximity limit stops the robot whatever Jev reports, a late (> 20 ms) or missing answer stops it, and an uncertain answer counts at its upper bound. Real hardware waits for a care partner, an ethics committee and a CE/MDR path.

| Results, limitations and status

Evidence at this revision

- Five G4-search descendants passed two private 30-seed navigation comparisons with retained lineage. This supports navigation promotion only.
- The proposed 10,000-capability atlas is mainly scaffold: ten draft packets are not ten thousand trainable expert capabilities.
- The original architectural and public-data figures in chapters 1–6 document the earlier v0.1 prototype; current production readiness must be checked against the code and fresh deployment tests.
- The business, PLS/PAR and physical paths in chapters 8–29 are proposed gates and scenarios, not observed commercial outcomes.

Limitations

- The mapping from infant brains is an analogy; the study is correlational, single-cohort and needs replication, and so does every structural claim about Toddler.
- Reward and utility weights are documented starting points, not calibrated values.
- The structure module is validated on social-network data, not yet on Toddler's own versions; that needs many recorded versions first.
- Competing with large robotics programmes on scale is not a goal; the contribution is a governed, auditable base for fine-tuning open models.

FOR DEVELOPERS

Start from `toddler-config/v1` : define the role, add your company's guardrails, and register experts only with measured evidence. Run `python3 -m pytest -q` before every deployment.

A trained Toddler is a candidate, not a worker

The route to useful work starts with a narrow, reproducible capability. A generation label means that a candidate survived a defined trial; it does not certify broad intelligence.

As of 11 October 2026, five G4-search descendants have passed two private 30-seed navigation trials against G3, with lineage retained. That is evidence for the tested navigation distribution. It is not evidence that they can reason across disciplines, operate a robot, or earn revenue. The capability atlas is also incomplete: ten draft packets and thousands of scaffold entries are an index to future work, not ten thousand mastered skills.

The first contract

For each capability, record the input distribution, permitted tools, expected output, time limit, STOP conditions, energy budget, and a human-readable failure taxonomy. A Montessori-style unlock-and-pick-up task, for example, separates perception, plan choice, manipulation, and recovery. This makes it possible to tell whether a child learned the task or merely memorised a scene.

Every child carries its ancestor, code and model hashes, training provenance, evaluation set identifier, and rollback image. Promotion requires a locked private test, paired comparison with the incumbent, retention tests, and a reported confidence interval. A failed child remains useful as evidence but receives no deployment permission.

PROMOTION UNIT

Promote one measured capability and one deployment scope at a time. No score on a toy task grants open-ended tool or physical authority.

| Curriculum: from reflex to transfer

Training should move from reliable small acts to combinations, then to unfamiliar variants where a prompt shortcut no longer works.

Four learning loops

1. Acquire: supervised examples and safe exploration teach one skill, such as counting objects or opening a box.
2. Compose: the Toddler combines skills: read instructions, count parts, plan a sequence, and verify completion.
3. Transfer: test new languages, layouts, textures, tools, and distractors that were absent from training.
4. Retain: replay older tasks after every update and measure catastrophic forgetting.

The teacher assigns tasks adaptively, but never evaluates against the same items used by the prompt improver. It can choose exercises from the public curriculum; an independent evaluator owns the sealed promotion set. Explanations, trajectories, and corrections are stored with provenance and licence. Ground truth or reproducible execution is preferred to a free-form model judge; ambiguous cases receive blinded human review.

Plasticity is controlled rather than unlimited: a candidate may update adapters, memory, router weights, or prompts, but must name the changed component and show its effect in an ablation. A fast gain that erases safety behaviour or older skills is a regression.

GRADUATION

Pass known tasks, novel variants, retention, latency, cost, and STOP tests separately; a single composite cannot hide a dangerous failure.

| From capability packets to agents

An agent is a governed workflow wrapped around capabilities: a task contract, tools, memory, permission boundaries, and an evaluator.

A packet in the atlas should contain theory, worked examples, expert practice, executable exercises, references, licences, and a small map of prerequisite skills. A code graph links a skill to repositories and the functions that implement it. Embeddings and t-SNE can help people browse related packets; neither supplies proof that the Toddler understands them. LightRAG can retrieve evidence, with citations and freshness recorded.

Agent assembly

Start with one workflow, such as a software debugger. Give it read access to a repository, a disposable test environment, a bounded time budget, and a patch proposal tool. Add a reviewer and a teacher that can veto unsupported claims. The router may call a small fast model, a specialist, or a stronger teacher, but every call is logged with model version, input digest, output digest, latency and measured energy. Tool grants are explicit and revocable.

The first agent trial is shadow mode: it proposes an action while a person or incumbent performs the task. Next come supervised execution and narrowly scoped autonomy. A promotion board compares quality, speed, energy, security findings, and human correction effort on a held-out task distribution.

DEFINITION

A useful agent completes an end-to-end task with verifiable output and bounded intervention, not merely a plausible answer.

| Mental work: software and analysis

Digital work is the first deployment lane because outputs can be replayed, diffed, tested and rolled back.

The initial occupations are coder, reviewer, architect, debugger, data analyst, and research assistant. Each needs its own private task bank and outcome measure. A coder is scored on build and behavioural tests plus maintainability; a reviewer on confirmed findings, false positives and severity; an architect on constraints satisfied after implementation; a debugger on root cause and regression prevention. Counting one snapshot of review findings does not establish who is the better software maker.

A Toddler + Teacher + specialist agent on ClaudeClaw should be compared with Toddler + Teacher and with a strong single-model baseline on the same private items. Record tool calls, tokens, elapsed time, GPU-board Wh, human minutes, and correctness. Public benchmarks are historical reference only; they do not confer promotion. Where answers lack an exact oracle, blinded independent raters use a preregistered rubric and adjudicate disagreements.

Begin with low-impact tasks: test repair, documentation from source, data quality checks, and sandboxed feature branches. Promote to production changes only after security and rollback gates. Business-plan drafting remains an advisory workflow: assumptions, citations and spreadsheet arithmetic must be visible to a human owner.

WORK PRODUCT

Sell reviewed outcomes and saved human time. Do not count generated text or closed tickets without acceptance evidence.

Physical work: embodiment changes the gate

A controller that can answer questions is not yet a safe robot operator. Real-world action requires a separate safety case.

Begin in simulation with a bounded object-handling task. Collect perception errors, control latency, collisions, near misses, and recovery traces. Then use a supervised workcell with guards, emergency stop, speed and force limits, geofencing, and a trained on-site operator. The language model proposes goals; a deterministic controller and independent safety layer enforce limits. Sensor loss, stale state, uncertain object identity, or a missed deadline causes a safe stop.

Transfer from simulation to reality is a measured gap. Change lighting, clutter, object mass, camera pose and human proximity one variable at a time. Re-test after every model, firmware, gripper or site change. Industrial safety standards and the applicable conformity route must be assessed by qualified engineers for the actual machine and use case. Human-like appearance is irrelevant to the safety claim.

Research such as SayCan and RT-2 shows viable ways to connect language, affordances and robot actions; those publications are background, not evidence for this Toddler deployment. A first useful physical task is repetitive, reversible and observable, such as sorting harmless parts in a partner-owned cell.

PHYSICAL PROMOTION

No autonomous customer site until independent hazard analysis, supervised pilot results, incident response and insurance are in place.

| The operating loop and human role

The path scales when learning, deployment, inspection and rollback form one observable loop.

A customer states an outcome and constraints. The planner decomposes it into auditable jobs. The router chooses a model or agent using a quality, latency and energy frontier measured on the current hardware. A worker acts through scoped tools. An independent verifier checks artifacts, tests or sensor evidence. A human accepts the outcome when stakes or uncertainty require it. Each stage leaves an event in the audit chain.

Three levels of authority

- Advice: the system explains and proposes; a person acts.
- Supervised execution: the system acts in a sandbox or guarded cell; a person can stop or roll back.
- Bounded autonomy: only pre-authorised low-impact actions with automatic monitoring and expiry.

The teacher reviews plans, evidence and failure clusters, but cannot silently approve its own candidates. Ancestors act as frozen comparators and sources of recovery policies. External contributors may submit descendants with public or private model provenance, reproducible trial descriptions and a pull request. A private model need not expose weights, but its role, version, costs and evaluation traces must be disclosed enough for the claim to be checked.

HUMAN CONTRIBUTION

People set goals, define acceptable harm, adjudicate ambiguity, and own the consequences of deployment.

Measurement that survives contact with reality

A benchmark matters only when it predicts accepted work outside the test room.

Keep a sealed, versioned evaluation bank owned by an independent evaluator. Separate development tasks, canaries, and promotion tests; rotate private items and scan candidate training data for overlap. Record prompts and decode settings so quants and mixtures are comparable. Add task-level uncertainty intervals, not merely a weighted average. For a router, compute the oracle ceiling from per-item results before training it; compare with the best single model and a random router.

Use a prospective customer pilot to check external validity: whether benchmark rank predicts accepted artifacts, fewer human minutes, lower incident rate and lower energy per accepted answer. Pre-register the primary outcome and threshold. A failed pilot means the benchmark or agent needs revision; it is not repaired by choosing another metric afterward.

Quality per energy uses measured GPU-board Wh plus a separately stated whole-system estimate. A slow expert may improve correctness but worsen cost and throughput; include retries and verification calls. Physical work additionally tracks successful cycles per supervised hour and near misses. A score of 100% on a small specialist suite is a description of that sample, never a claim of perfect competence.

EVIDENCE CHAIN

Private task → immutable run → independent verdict → customer acceptance → model or agent promotion.

| A practical sequence of releases

Do the smallest useful work first, then widen scope only when a measured result earns it.

Gate	Deliverable	Proof required
0	Reproducible G4 registry and private trials	Lineage, hashes, paired gains, retention
1	One digital specialist in shadow mode	Sealed tasks and accepted outputs
2	Teacher-reviewed digital work	Quality, human minutes, cost per acceptance
3	Multi-agent customer pilot	Prospective outcome and incident log
4	Guarded physical cell	Safety case and supervised cycles

The first 90 days should freeze a repeatable baseline, select one software workflow, instrument energy and review effort, recruit two design partners, and run shadow comparisons. The next 90 days can add a second digital specialist only if the first meets its gate. Physical pilots begin with a partner that owns the workcell and can supply engineering supervision.

Stop or narrow the programme when accepted-work quality is below the incumbent, human correction erases speed gains, or safety evidence is missing. A broad “human-level” claim is never a milestone. The operational milestone is a customer who repeatedly accepts a bounded result at a disclosed cost and risk.

NORTH STAR

Accepted, independently verified work per euro, joule and human review minute.

| Two ledgers, two kinds of evidence

In the Pulse design, PLS and PAR are distinct accounting mechanisms. Their proposed Toddler integration is a design, not a live production economy.

PLS is the Pulse pay token associated with verified useful computation. A job has a request, budget, executable specification, output, verifier and settlement decision. PAR is the Pulse augmented-reality observation coin associated with a confirmed field observation linked to a funded bounty. The PAR ledger records digests and a pod reference, not raw camera content. Their supply and reward policies are separate; one token is not an interchangeable proxy for the other.

A Toddler agent can propose a computation job or a field-observation request. It cannot certify its own output or mint a reward merely by writing a plausible report. The verifier and treasury paths must accept the result under their own rules. For a robot, the act of moving an object and the evidence that it moved are also separate. A controller trace may support an audit but does not by itself prove that the intended world state changed.

PLS and PAR are described in local Pulse draft papers and code inspected on 11 October 2026. Token issuance, adoption and legal classification remain unresolved dependencies; revenue forecasts below exclude token appreciation and sales.

DESIGN RULE

Reward evidence of useful work; never reward an agent merely for producing more agent activity.

| PLS: verifiable digital work

PLS could coordinate distributed compute when a job can be specified, verified and priced before execution.

The cleanest examples are deterministic transforms, test execution, document conversion with exact output digests, and reproducible data processing. A proposer records the input digest, container or code digest, resource ceiling and expected verifier. A worker returns the artifact and proof bundle. Sampled re-execution and collateral make false results costly; settlement follows only after the verifier accepts. A Toddler may route a job to a peer if measured expected cost, latency and risk beat local execution.

LLM text is harder: two valid answers may have different bytes, and a model judge can be wrong or compromised. For such jobs, acceptance needs a task-specific oracle, validated artifact, or human sign-off. Nondeterministic GPU kernels may need tolerances or hardware attestation; the Pulse proof-of-useful-work document notes that tolerance digests are not wired through every job path. The pilot must use currently implemented verification, not assume future mechanisms are active.

The Toddler integration point is a narrow job adapter: submit, monitor, verify, settle, and write an audit event. It should never expose wallet keys to a prompt. Set budgets in euros and joules even if a later settlement uses PLS, so model routing stays economically intelligible.

FIRST PILOT

Use a reproducible batch job with known output, compare local and peer cost, and test rejection of a deliberately corrupted result.

| PAR: field observations and robots

PAR addresses an evidence gap: useful physical work often depends on observations that a digital verifier cannot replay.

A funded observation bounty names a place, time window, requested signal, permitted sensor, privacy constraints and confirmation rule. A device captures evidence, stores raw material under controlled retention in a pod, and submits digests and metadata. Independent checks decide whether the observation meets the bounty. A confirmed event may trigger PAR accounting; unconfirmed, duplicate or out-of-scope events do not.

For a Toddler-guided robot, the observation layer can verify that a bin is empty or a part is present before and after a bounded action. It can also reveal disagreement between the agent's plan and the world. Yet cameras can be spoofed, scenes staged and sensors miscalibrated. Multi-view evidence, timestamps, device attestation where available, human sampling and adversarial trials are required before a claim of physical completion.

Minimise personal data: keep faces, speech and raw video out of the public ledger. Consent, retention and local access controls must be designed for the site. The current Pulse observation code's confirmation gate is a software condition, not proof of truth in the physical world.

FIRST PILOT

Check a harmless, repeatable cell state with two independent sensors and blinded manual audits before linking PAR to rewards.

Joining the digital and physical loops

A useful robot job can combine PLS-backed computation with PAR-backed observation, while keeping their proofs distinct.

Stage	Evidence	Decision owner
Plan	Task contract, site limits, operator approval	Customer
Compute	Code, inputs, output digest, replay or test	PLS verifier
Act	Controller trace and safety interlocks	Robot safety layer
Observe	Sensor digest, pod reference, confirmation	PAR verifier
Accept	Business outcome and incident record	Customer and auditor

For example, an agent plans a small inventory count. Distributed compute classifies images under a reproducible model version. A supervised robot scans shelves. Separate observations verify location and stock count. The customer accepts the inventory file after discrepancy review. PLS could settle the compute job; PAR could recognise confirmed observations. Neither token should bypass the customer acceptance gate.

Use a common event ID and cryptographic references to join records, while access control keeps private material off public infrastructure. Reconciliation must tolerate reorganisation, delayed evidence and disputed observations. If a verifier is unavailable, the work pauses or follows a documented human exception; it does not silently mark itself complete.

INTEGRATION TEST

Trace one job end to end, then inject a false compute result and a false sensor report; both must be rejected independently.

| Economic and governance boundaries

A token mechanism is useful only if it improves verification or coordination after all settlement, legal and operational costs.

Before any live token pilot, compare three routes on the same work: local internal accounting, conventional euro billing, and PLS/PAR settlement. Measure verification expense, fraud loss, delay, privacy exposure and net customer value. Token rewards must have a funded demand source and capped emission policy; promotional minting would contaminate the unit economics. A customer should be able to buy a verified service in euros without acquiring a speculative asset.

Pulse's PLS and PAR papers are drafts. Their code paths provide design evidence, not legal clearance or audited deployment. Dutch and EU counsel should assess token classification, consumer terms, tax, data protection and the applicable MiCA regime for a concrete implementation. Custody, key rotation, dispute resolution and chain failures need an incident runbook. A robot incident is handled under operational safety procedures regardless of token settlement.

The Toddler business forecast therefore counts subscription, verified task and workcell service fees only. It gives PLS and PAR zero revenue until customer demand, verification reliability, compliance and an actual integration are demonstrated. The tokens can still be valuable as coordination experiments even if euro billing remains the commercial route.

GO/NO-GO

Proceed only when measured net value exceeds the simpler billing route without weakening audit, privacy or safety.

| Sell dependable work, not a human illusion

The proposed company is a privacy-first work platform built around Toddler, Teacher and specialist agents. Its commercial promise is a bounded result with an audit trail.

The entry customer is an organisation that already has a backlog of repetitive software, data-quality or research tasks and cannot send all inputs to an external API. Its pain is not a shortage of generated prose; it is the cost of reviewing, reproducing and securing outcomes. The first sale is an annual orchestration subscription. Additional independently verified tasks are metered. Physical work is added as a software-and-service fee for customer- or partner-owned cells only after a supervised safety pilot.

The world may develop much more capable agents and robots during these five years. The base forecast does *not* assume that Toddler reaches human parity. It requires a series of narrow workflows that save measurable customer time. Robot-market adoption is context, not an addressable-market multiplier. IFR reported 542,076 industrial robot installations worldwide in 2024; that number does not imply demand for this product.

Customers buy in euros. PLS and PAR may later coordinate verifiable compute and observations, but speculative token value is excluded from the plan. The scarce asset is a trusted dataset of task attempts, corrections, tests and accepted work, governed by customer consent and provenance.

CUSTOMER PROMISE

A result that can be inspected, repeated or reversed, with clear human accountability.

| Three products, one evidence layer

The product line widens only as each layer earns trust in deployment.

Offer	Customer receives	Commercial unit
Digital workbench	Scoped agents, private evaluation, audit and routing	€24,000 per organisation-year
Verified tasks	Accepted output with evidence and replay or review	€0.12 net fee per billable task
Physical cell service	Workflow software, monitoring and incident support	€72,000 per partner-owned cell-year

The workbench includes orchestration and a contractual allowance of ordinary requests; the task fee applies only to separately billable verified jobs after that allowance. A billable task must have a pre-agreed acceptance rule, so internal model calls and retries are costs, not new sales. The cell fee excludes robot purchase or lease, integration of unsafe machinery, and speculative token rewards.

A shared evidence layer records provenance, permission, model routing, verifier results, energy, correction effort and customer acceptance. This lets the company compare Toddler-only, Toddler + Teacher, and Toddler + Teacher + specialist workflows on exactly the same tasks. The layer is also the customer-facing assurance product; agents and model mixtures can change without erasing history.

PACKAGING TEST

Before launch, two design partners must confirm that the offer saves more accepted-work cost than its subscription and review burden.

| Go to market through narrow pilots

Start with workflows that have objective acceptance, frequent repetition, and a clear budget owner.

Year one targets eight paying organisations, reached through two design-partner pilots and a controlled expansion to software teams, data operations and privacy-sensitive research groups. Every pilot has a baseline period, a shadow period, and a decision gate. The buyer sees accepted-task rate, human review minutes, incident count, turnaround time and fully allocated cost. Reference cases must disclose the task distribution and denominator.

Distribution can combine direct technical sales, open-source contributions, and system integrators. Crowdsourced Toddler descendants are welcome through pull requests carrying lineage and trial evidence, but commercial deployments use a reviewed release channel. A partner may run its own private model; the contract still requires the version, licence constraints and evaluation method to be stated. The platform does not promise that a contributor's private data becomes public.

Physical expansion begins with a robot integrator and a single guarded site. Industrial and service robotics are already active markets, yet each cell needs site-specific economics and engineering sign-off. Do not count a workcell as active until it completes a paid supervised month and meets the safety and customer acceptance gate.

SALES DISCIPLINE

Convert a pilot only when the customer can name the accepted outcome, its previous cost, and the person accountable for the new process.

| Five-year operating assumptions

These are illustrative planning inputs in nominal euros, excluding VAT, for 2027–2031. They are not measured demand or booked revenue.

Year	Paying orgs	Billable verified tasks	Partner-owned cells
2027	8	0.2m	0
2028	25	1.0m	2
2029	70	4.0m	8
2030	160	12.0m	25
2031	320	30.0m	70

Prices are held flat at €24,000 per organisation-year, €0.12 per billable verified task, and €72,000 per active cell-year. These are modelling choices to be tested in customer discovery, not market quotes. The forecast assumes each listed customer or cell contributes a full year; a real sales ramp would reduce early-year revenue unless volumes or prices compensate. This makes the plan a target run-rate scenario rather than a probability-weighted budget.

The task figures exclude internal retries and free/included requests. Any task that fails independent acceptance contributes delivery cost but no fee. The forecast assumes the listed billable tasks already passed that gate; actual conversion must be measured.

REPRODUCTION

All arithmetic comes from `scripts/forecast_toddler_business.py` and its JSON output in this paper's directory.

| Unit economics before scale

The plan is viable only if accepted-work margins survive retries, verification and customer support.

Model direct delivery cost as 25% of workbench revenue, 40% of verified-task fees, and 35% of cell-service fees. These shares include inference, GPU electricity, verification, storage and direct service labour. They are hypotheses. Meter whole-system kWh per accepted task and distinguish it from GPU-board Wh; a night queue does not make incremental power free. For reference, Eurostat reported an EU non-household average electricity price of €0.1837/kWh for the second half of 2025; a specific site may differ materially.

At €0.12 per verified task, a 40% direct-cost share leaves €0.072 gross contribution. Ten million accepted, separately billable tasks would therefore contribute €720,000 before operating expense. If an average task needs repeated expensive teacher calls, the 40% assumption fails. The gate is not token throughput but net contribution per accepted result, measured prospectively.

At €24,000 per customer-year, a 25% direct-cost share leaves €18,000 before sales, research and administration. At €72,000 per active cell-year, a 35% direct-cost share leaves €46,800. Cells require a separate allowance for field support and incident response; a partner owns the machine and ordinary hardware depreciation under this model.

UNIT GATE

Do not scale a workflow with negative gross contribution after retries and verification, even when its benchmark score is excellent.

| Base-case income path, 2027–2031

Revenue is organisations $\times$ €24,000 + verified tasks $\times$ €0.12 + cells $\times$ €72,000. Direct costs and operating expenses follow the stated assumptions.

€ million	2027	2028	2029	2030	2031
Revenue	0.216	0.864	2.736	7.080	16.320
Direct costs	0.058	0.248	0.814	2.166	5.124
Gross profit	0.158	0.616	1.922	4.914	11.196
Operating expense	0.850	1.500	2.800	5.100	8.500
EBITDA	-0.692	-0.884	-0.878	-0.186	2.696

Operating expense includes product engineering, sales, trust and safety, and general administration; the model does not split their headcount yet. A credible operating budget must replace these envelopes with hiring plans, salaries and contracts before financing. Base-case EBITDA turns positive only in 2031. This does not mean profit or positive cumulative cash flow: depreciation, tax, interest, working capital and financing are not modelled.

The output is sensitive to customer and task volume. A company should run the same formula with observed conversion and actual power, review and support costs after each pilot.

INTERPRETATION

These numbers express the required scale for a business case; they do not predict that Toddler will achieve it.

| Cash need and financing gates

The plan consumes cash for several years before the modelled 2031 operating turn.

€ million	2027	2028	2029	2030	2031
Capital expenditure	0.180	0.300	0.450	0.750	1.200
EBITDA less capex	-0.872	-1.184	-1.328	-0.936	1.496
Cumulative proxy	-0.872	-2.056	-3.384	-4.320	-2.824

The deepest base-case cumulative cash proxy is €4.320m at the end of 2030. Adding a 25% contingency gives a preliminary funding envelope of about €5.40m. This is *not* a complete financing requirement: working capital, taxes, interest, timing of collections, debt service and unexpected safety or legal costs are excluded. A monthly cash-flow model and pilot evidence are required before a financing decision.

Stage spending against proof: seed the evaluation and first customer pilots; fund product expansion only after paid accepted work; fund robot integration only after a supervised safety case and partner contract. Avoid buying a robot fleet in the base plan. Partner- or customer-owned equipment keeps capital tied to evidence instead of speculation.

FUNDING RULE

Release the next tranche against accepted-work and safety gates, not model-score headlines or token price.

| Downside, upside and falsification

A forecast should tell us what would disprove the plan soon enough to change it.

The downside case halves paying organisations, billable tasks and active cells while retaining base operating and capital spending. In 2031 it yields €8.160m revenue and **-€2.902m EBITDA**; its five-year cumulative cash proxy is -€12.259m. The upside case scales those volumes by 1.5 and operating expense by 1.2, retaining capital spending; 2031 revenue becomes €24.480m and EBITDA €6.594m. These are arithmetic sensitivities, not likelihood estimates.

Test the most uncertain assumptions first: customer conversion, the share of tasks independently accepted, human review minutes, direct cost per retry, and the time needed to certify a physical cell. One quarter of weak conversions should trigger pricing and product interviews, not a bigger sales target. If a teacher improves quality but uses too much energy or reviewer time, route it only to tasks where the gain pays for itself.

A live dashboard should compare actuals with this model every month. Show forecast error by assumption, not just total revenue. Keep published benchmark results, work acceptance and sales claims separate. Early pilots may invalidate the 2031 scale and still reveal a valuable smaller company.

DECISION THRESHOLD

After two paid pilots, revise the model from measured conversion, acceptance and cost. If no repeatable positive gross contribution exists, narrow or stop the offer.

| Five years of responsible execution

The age of more capable agents and robots increases the value of human choice, verifiable work and safe deployment.

2027: prove one digital workflow with sealed tests and eight paying organisations. 2028: add two supervised physical cells only if the safety case passes. 2029: standardise the evidence layer and partner integration across seventy organisations. 2030: test whether wider routing and specialist portfolios improve accepted-work economics at 160 organisations. 2031: reach sustained service quality at the modelled 320 organisations and seventy cells, or publish why the path was revised.

Workers and customers retain decision rights. Deployments should document what tasks are automated, which new supervisory work appears, how errors are appealed, and who may stop the system. A more human-like conversation interface should never be confused with consciousness, competence or legal personhood. The company should invest in training operators and reviewers as deliberately as in models.

PAR and PLS can be tested as optional evidence and coordination mechanisms after operational and legal gates. A robot partnership should be allowed to use ordinary euro invoices and independent logs. The company succeeds if it delivers accepted outcomes with accountable human governance, whether or not agents ever equal people broadly.

FIVE-YEAR MEASURE

Repeatable customer acceptance, positive unit economics, safe physical operations and independently auditable claims.

Appendix

A. References

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B. Glossary

Clustering coefficient - the share of a node's neighbours that are connected to each other (0 to 1).

Consensus clustering - repeating a non-deterministic clustering many times and clustering the co-assignment matrix.

Jev - a model that answers many typed questions in parallel and returns probabilities with uncertainty; used as a feature generator, never as the sole authority.

Pulse - knitweb's trustless relay to compute and node-to-node resource sharing.

PLS - Pulse pay token proposed for independently verified useful computation.

PAR - Pulse augmented-reality observation coin proposed for confirmed field observations linked to a bounty.

Cash proxy - EBITDA minus capex in this paper; it excludes several real cash-flow items and is not free cash flow.

STOP rule - a hard constraint that vetoes an action before utility is compared.